Combinatorics 5

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Problem 1

The posets with underlying set $\{a, b, c, d\}$ are:

1. No relations,
2. $a \leq b$,
3. $b \leq a$, $c \leq a$,
4. $a \leq b$, $a \leq c$,
5. $a \leq b$, $c \leq d$,
6. $b \leq a$, $c \leq a$, $d \leq a$,
7. $a \leq b$, $a \leq c$, $a \leq d$,
8. $a \leq c$, $a \leq d$, $b \leq c$, $b \leq d$,
9. $a \leq b$, $b \leq c$,
10. $a \leq b$, $a \leq c$, $c \leq d$,
11. $a \leq b$, $c \leq b$, $d \leq c$,
12. $a \leq b$, $a \leq c$, $b \leq d$,
13. $b \leq a$, $c \leq a$, $d \leq b$,
14. $a \leq b$, $a \leq c$, $b \leq d$, $c \leq d$,
15. $a \leq b$, $b \leq c$, $c \leq d$,
16. $a \leq b$, $c \leq b$, $c \leq d$.

Problem 2

Let the set be $S = \{a, b, c, d, e, f\}$. From the Hasse diagram's edges, we have that

$$b \geq d, \quad b \geq e,$$
$$c \geq e, \quad c \geq f,$$
$$a \geq d, \quad a \geq f.$$

Our goal is to find $f : S \to [6]$ with $f(x) \leq f(y)$ if $x \leq y$ for all $x, y \in S$. Since a, b, and c are each $\geq$ two other elements, we have that $\{a, b, c\} \subseteq \{3, 4, 5, 6\}$. We can split this into two cases:

Case 1: assume $\{a, b, c\} = \{4, 5, 6\}$. Then $\{d, e, f\} = \{1, 2, 3\}$, and choices of a, b, c are independent from choices of d, e, f, so the number of ways is $(3!)(3!) = 36$.

Case 2: assume $3 \in \{a, b, c\}$. First, choose either a, b, or c to be 3 (assume it's a WLOG, can be done in 3 ways). Since $a \geq d$ and $a \geq f$, $\{d, f\} = \{1, 2\}$, so d and f can be chosen in 2 ways. Then, since $b \geq e$ and $c \geq e$, $e = 4$ and $\{b, c\} = \{5, 6\}$. So b and c can be chosen in 2 ways. The result is $3 \cdot 2 \cdot 2 = 12$.

Adding the two results, the total count is 48.

Problem 4

Given arbitrary $N \in \mathbb{N}$, we can construct an antichain of size $\geq N$. If N is even, consider the points:

$$S = \{(i, N - i + 1, 1, \ldots, 1) : i \in [N]\}.$$

Then, given any two $x, y \in S$, if $x[0] \leq y[0]$ then $x[1] \geq y[1]$ and $x[0] \neq x[1]$ and $y[0] \neq y[1]$. So $x \not\leq y$ and $y \not\leq x$, and S is an antichain of size N. If N is odd, construct an antichain of size $N + 1$ in this way.

To show there is no infinite antichain, we use induction on k. If $k = 1$, then $\mathbb{N}$ is totally ordered so there is no antichain of size > 1.

Now, assume true for $k - 1$. Take any infinite sequence $v^{(1)}, v^{(2)}, \ldots$ with each $v^{(i)} = (a_1^{(i)}, \ldots, a_k^{(i)})$. From the first coordinates $a_1^{(1)}, a_1^{(2)}, \ldots$, select a nondecreasing subsequence $a_1^{(i_1)}, a_1^{(i_2)}, \ldots$ and take the corresponding vectors $v^{(i_1)}, v^{(i_2)}, \ldots$. Define $w^{(i)}$ as the last $k - 1$ items in $v^{(i)}$. By the induction hypothesis, among the $w^{(i)}$, there exist i, j with $w^{(i)} \leq w^{(j)}$. Together with the nondecreasing first coordinates, we get $v^{(i)} \leq v^{(j)}$.

Problem 9

Define $A_n = \{n, n+1, \ldots\}$ and the chain $C = \{A_n : n \in \mathbb{N}\} \cup \{\varnothing\}$. This is a chain since $A_1 \supseteq A_2 \supseteq \cdots \supseteq \varnothing$, and it's countable. To show that this is maximal, let $S \subseteq \mathbb{N}$ be comparable with every set in C. If $S = \varnothing$ we're done, so assume $S \neq \varnothing$ and let $m = \min(S)$. Then $S \subseteq A_m$ but $S \not\subseteq A_{m+1}$. So we must have $A_{m+1} \subseteq S$. Therefore, S contains all integers $> m$, so $S = A_m$ and $S \in C$. So no new subset can be added while keeping it a valid chain, and therefore C is maximal.

Problem 10

The only possible end result is where the a6 pawn promotes to a knight, goes to c8, and the rook goes to a8, allowing Bg2#. The minimum number of pawn moves required is 5, the minimum number of knight moves is 5, and the minimum number of rook moves is 2.

Now, we count the number of ways to move a knight from a1 to c8. Working backward, the knight can come from either d6 or e7, so the number of ways from each of those squares is 1. Then, we can move to d6 from b5, f7, or f5, and we can move to e7 from d5, f5, or g7, so the counts for each of those is 1, except for f5 which is $1 + 1 = 2$. Continuing to backtrack (and limiting the path length to 5), we get that the number of knight paths from a1 to c8 is 10.

Finally, we can place any two moves of certain pieces between the required pawn, knight, and rook moves. At this point, I had trouble figuring out how to incorporate every possible extra movement of the rook, pawns, and knight into the total count.

Problem 15

The minimum value of $r(\mathcal{A})$ is $m + 1$, since every added distinct hyperplane increases the number of regions by at least 1 (it splits at least one existing region), and this can be achieved using parallel planes.